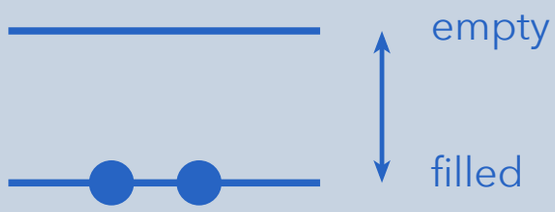
A calculation, a charging experiment, and a light beam ask different questions.

Comparing the right gaps is essential when predicting conductivity, solar absorption, or a material's color.

1 READ THE ORBITALS

Kohn–Sham (KS) gap: auxiliary levels

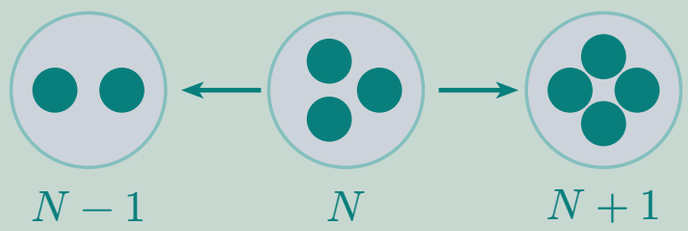


$$E_{\mathrm{KS}} = \varepsilon_{\mathrm{L}} - \varepsilon_{\mathrm{H}}$$

H: highest occupied; L: lowest unoccupied.

2 ADD / REMOVE CHARGE

Fundamental gap: many-electron energies

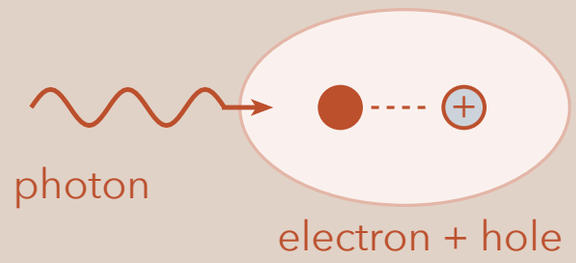


$$E_{\mathrm{g}} = I - A$$

I: removal cost. A: energy gained on addition.
N: number of electrons.

3 SHINE LIGHT

Optical gap: a neutral excitation

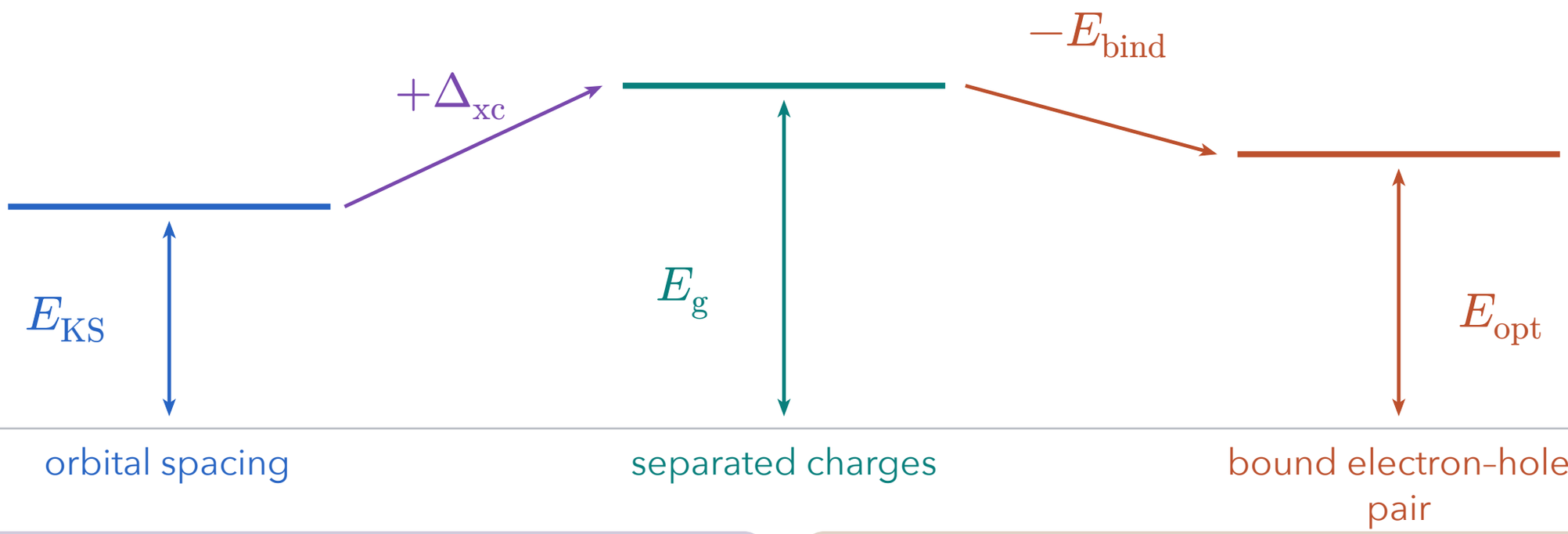


$$E_{\mathrm{opt}} = h\nu_{\mathrm{onset}}$$

$h\nu$: photon energy; ν: frequency. A hole is a missing electron in a filled state.

Two corrections, two different pieces of physics

Charging changes the electron count, so Δ_{xc} converts the KS orbital gap into the fundamental gap.
Light keeps the count fixed: electron–hole attraction lowers the excitation energy by E_{bind}.
Shown: a direct gap (no momentum from lattice vibrations needed) and a bright exciton (a bound pair that light can create). Not to scale: the two corrections compete, so the optical gap can lie above or below the KS gap.



$$E_{\mathrm{g}} = E_{\mathrm{KS}} + \Delta_{\mathrm{xc}}$$

Derivative discontinuity: the exact exchange-correlation (XC) potential jumps at an integer electron count. Even exact KS orbital energies need this correction.

$$E_{\mathrm{opt}} = E_{\mathrm{g}} - E_{\mathrm{bind}}$$

Exciton binding: electron–hole attraction lowers the neutral excitation below the matching free-pair threshold. An exciton is this bound pair.

$E_{\mathrm{g}} = E(N+1) + E(N-1) - 2E(N)$, where $E(N)$ is the ground-state total energy at fixed nuclei.

Read the conditions: indirect optical transitions can need phonons for momentum; selection rules can make low excitations dark. The relation above uses a matching direct threshold.

DFT takeaway: identify the observable before comparing gaps. The KS relation assumes a local multiplicative potential; hybrid generalized-KS gaps require their own interpretation.